

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Designed, built, and deployed a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** in a mission-critical workflow via REST integrations connecting external systems; contributed to **\$340K+** in revenue.
- Built a full-stack platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Followed secure programming best practices to keep the system reliable, adding integration tests, auth hardening, rate limiting, and AWS KMS encryption, and remediated **20+ VAPT findings**.
- Served as primary technical POC, troubleshooting, analyzing, and resolving production incidents to root cause, collaborating across teams, and piloting a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with object-oriented design, input validation, and clear, backward-compatible contracts used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production feature running a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation and short-circuit caching.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with REST APIs, a multi-branch data model, and cron automation, maintained end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a backend service exposing **REST API endpoints** with typed request planning, deterministic extraction, and provider routing; hardened with **150+ tests** and **Docker** deployment to keep it reliable under changing inputs.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a distributed, **event-driven** service with durable, multi-step workflows that survive process restarts and tool failures, streaming results to the client over WebSocket in **real time**.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a high-volume data pipeline over a **~1.4B-row** dataset with broadcast joins, caching, partitioning, and skew handling, cutting end-to-end runtime **~25%**, profiled via Spark UI.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Java, C++, Python, TypeScript, JavaScript, SQL, Bash

Backend & APIs: REST APIs, object-oriented design, event-driven systems, messaging (Kafka), caching (Redis), concurrency

Databases: PostgreSQL, MongoDB, Redis, SQL, SQLite

Testing & DevOps: Integration testing, automated testing, CI/CD, Docker, Kubernetes, Git, GitHub Actions

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.